Smoke Point In Coflow Experiment
"SPICE"
Microgravity Science Glovebox
Increment 19-20 (30 Day Report)

Investigators: David Urban NASA GRC, Peter Sunderland: Univ. Maryland
Project Scientist: Zengguang Yuan, National Center for Space Exploration Research
Project Manager: Robert Hawersaat NASA GRC

I. Introduction

A. Hypothesis

The hypothesis of the SPICE investigation is that smoke point behavior will be observed for co-flowing flames and that it will show dependence on nozzle diameter and coflow velocity.

B. Objectives of Investigation

The objective of the SPICE experiment is to find the laminar smoke-point properties of nonbuoyant jet diffusion flames (i.e., the properties of the largest laminar jet diffusion flames that do not emit soot) for several fuels under different nozzle diameter/ co-flow velocity configurations. Luminous flame shape measurements would also be made to verify models of the flame shapes under co-flow conditions.

C. Previous Mission Experience

N/A

D. Background/History of Project *(Optional)*

The SPICE experiment was an outgrowth of the Laminar Soot Processes Experiment (LSP) which flew on STS 83, 94, and 107. SPICE utilized the design heritage and modified the hardware from the Enclosed Laminar Flames Experiment (ELF) which flew in the Middeck Glovebox on STS 87.

II. Methods/Research Operations

A. Discussion of Method/Protocol

Most tests were conducted by setting the coflow velocity, igniting the flame, and adjusting the fuel flow rate. Approximate smoke points were identified in the video by the flight crew and simultaneously by the ground support crew in Cleveland. Typical burn times were 60 s, and after each two such burns the MSG contents were flushed with ISS cabin air. After the tests the video record was analyzed to identify smoke point conditions. The associated coflow and fuel flow rates were obtained from digital data recorded in the video. The smoke point lengths, LSP, were obtained from the video record and are the measured distance from the burner tip to the tip of the luminous flame on the flame axis.

III. Results

A. List of Preflight, In-flight, Postflight Anomalies

SPICE operated very successfully with the only significant anomaly being the accumulation of soot in the outlet screen which changed the flow behavior in the duct. This was corrected by a crew member using the vacuum to clean the outlet screen on June 15.

B. Completeness/Quality of Data

During Increment 18 SPICE conducted 108 test points completing its nominal test matrix. During Increments 19/20 SPICE conducted 155 additional test points the provided improved resolution of the smoke points and allowed us to explore further limits in the parameter space.

C. Status of Data Analysis: The data from the down linked video have been analyzed for flame length and smoke point flow rate. These have resulted in a meeting apper and a draft journal paper. More work will be conducted with analysis of the digital luminance data from the Nikon.

D. Preliminary Research Findings:

See attached paper below

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Laminar Smoke Points of Microgravity Coflow Diffusion Flames

K.T. Dotson[a], P.B. Sunderland[a], Z.–G. Yuan[b], D.L. Urban[c]

[a]Department of Fire Protection Engineering, University of Maryland, College Park, MD, USA
[b]National Center for Space Exploration Research, Cleveland, OH, USA
[c]NASA Glenn Research Center, Cleveland, OH, USA

Laminar smoke points were measured in nonbuoyant laminar jet diffusion flames in air coflow aboard the International Space Station as part of the Smoke Points in Coflow Experiment (SPICE). Smoke points were found for ethylene, propane, propylene, and mixtures of 50 and 75% propylene by volume in nitrogen. Burner diameters were 0.41, 0.76 and 1.6 mm and coflow velocities varied from 5.4 – 65 cm/s. These flames afford good control over residence time via variations in dilution, burner diameter, and coflow velocity. Increasing coflow velocity decreases luminous flame lengths, but moves smoke points to higher fuel flow rates and longer luminous flame lengths. Smaller burners have longer smoke points for coflowing flames. These trends can be explained qualitatively by considering total residence time effects. The luminous flame lengths generally correlate with fuel mass flow rate divided by stoichiometric mixture fraction, although they increase slightly as air velocity decreases.

1. Introduction

Soot formation is an important unresolved problem in combustion science. Soot is estimated to be responsible for up to 50,000 premature deaths in the U.S. [1] and is believed to contribute to climate change and glacial melting [2]. Radiation from soot in flames allows reading by candle-light but also contributes to heat loads in engines and the spread of unwanted fires. Smoking flames emit carbon monoxide, which is the major cause of death in fires [3].

The smoke point is the condition (or length) where a flame begins to emit soot. Smoke points indicate the sooting propensity of a fuel in diffusion flames. In normal gravity, laminar smoke points have been measured for many gaseous and condensed fuels [4–9]. Laminar smoke points have been found to correlate with soot volume fractions [10] and radiative loss factions [11] of turbulent diffusion flames. Yang et al. [12] found that gas turbine soot emissions correlate very well with a threshold sooting index (TSI) based on fuel molecular weight divided by the laminar smoke point. Smoke points have been shown to correlate with the smoke release from turbulent buoyant flames [13] and have been applied to developing general soot models for computational combustion simulations [14] and heat release measurements [15].

In a 1988 paper [16], Glassman predicted that smoke points would not occur in nonbuoyant gas-jet flames. He based this on the assumption that residence time scales with the square root of fuel flow rate in normal gravity but is independent of flow rate in microgravity flames. Subsequent work [17] observed smoke points of nonbuoyant flames aboard a parabolic aircraft and used numerical predictions to show that residence time increases with flame length for nonbuoyant flames without co-flow. Urban et al. [18] reported smoke points of microgravity flames observed on a space shuttle. Microgravity smoke point measurements are difficult in parabolic aircraft

owing to g-jitter and in drop facilities owing to limited test times.

Attributing smoke point behavior to total residence time helps with qualitative explanations of smoke point behavior but to date has not yielded successful quantitative correlations [8]. This may be because smoke points arise from a competition between the times available for soot formation and soot oxidation. Delichatsios and co-workers suggested that smoke point length is proportional to soot formation time [19–20]. Quantitative correlations may also be possible from an examination of radiative losses, which have been observed to be 20 – 30% for normal-gravity smoke point flames and to increase with adiabatic flame temperature [5].

Soot is transported in flames by convection and thermophoresis. By varying the burner diameter, coflow velocity, and fuel dilution, flame shape and residence times in the soot formation and soot oxidation regions can be changed. A decrease in coflow velocity will generally increase entrainment of flow towards the flame axis.

Kent and Wagner [21] found that smoke burn out in the oxidative region of the flame ceases when the tip of a buoyant diffusion flame cools to approximately 1300 K. Urban et al. [22] found that the tip of a nonbuoyant (microgravity) flame reaches 1000 K at its smoke point. In buoyant flames this decrease in temperature is attributed primarily to heat conduction, while in nonbuoyant flames it is attributed largely to radiative quenching [23]. The radiative quenching effect causes a visible tip opening in nonbuoyant smoke points of flames of relatively long residence time [22].

A recent 2.5 s drop tower study by Jeon et al. [24] found that for small coflow velocities the soot volume fraction increases with coflow velocity. This was attributed to an increase in temperature. Lin and Faeth [7,25] studied the effects of coflow velocity on weakly-buoyant flames near the smoke point. They observed flames in normal gravity, but at low pressures to reduce the effects of gravity. They found that increasing the coflow velocity reduced soot volume fractions and increased the smoke point length. This was attributed to decreased flame residence time and modified soot pathlines. Refs. [7 and 24] both showed that luminous flame lengths decrease with increasing coflow velocity. Their burner diameters, 7 and 10 mm, respectively, were much larger than those considered here.

Faeth and co-workers [25–27] applied the flame shape analyses of Spalding [28] and Mahalingam et al. [29] to nonbuoyant gas-jet flames to obtain

$$(L_f - L_o) / d = C_f \, Re \, Sc / 16 \, Z_{st} , \tag{1}$$

and

$$(L_f - L_o) / d = 3 \, C_f \, Re \, Sc / 32 \, Z_{st} , \tag{2}$$

for coflowing and non-coflowing cases, respectively, where L_f is luminous flame length, L_o is a virtual origin correction, d is burner diameter, C_f is an empirical coefficient, Re is the jet Reynolds number, Sc is the Schmidt number, and Z_{st} is the stoichiometric mixture fraction. The Reynolds number is defined as $\rho \, d \, u / \mu$, where ρ is density, d is burner diameter, u is velocity, and μ is dynamic viscosity. The Schmidt number and viscosity in Eqs. (1–2) were those of air at the mean of the adiabatic flame temperature and the ambient temperature.

Schmidt number and viscosity, defined in this fashion, do not vary substantially for the flames in this study, and are not considered significant. In addition the virtual origin correction was neglected. With this in mind, Eqs. (1–2) can be rearranged to

$$L_f = C\,\dot{m}_f / Z_{st}\,, \tag{3}$$

where C is a constant and $\dot{m}_f$ is the fuel mass flow rate.

Thus motivated, the objectives of this study are:

1. to measure nonbuoyant smoke points of propane, ethylene, propylene, and propylene/nitrogen mixtures,
2. to identify the effects of coflow velocity and burner diameter on the smoke points, and
3. to correlate the luminous flame lengths.

2. Experimental

The Smoke Points in Coflow Experiment (SPICE) module was brought to the International Space Station (ISS) in 2008. It was operated in the Microgravity Science Glovebox (MSG) by various crew members in 2009. Smoke points were attempted for all fuels for fuel jet diameters of 0.41, 0.76, and 1.6 mm. The conditions for which smoke points were found are summarized in Table 1. In some cases (e.g., tests involving ethylene) the fuel supply was depleted before desired smoke points could be found, as the supply was limited to two 75 cc bottles of each fuel at gage pressures of 760 – 1440 kPa, depending on the fuel.

The SPICE module contains a rectangular duct that has an approximately 76 mm square cross section with rounded corners. A DC fan supplies coflowing air from the glovebox contents. The air passes through a honeycomb and screen to reduce swirl and then enters the duct. The air and combustion products leave the duct through a perforated copper plate followed by a copper screen. The screen is 50 mm upstream of the burner discharge and the plate is 120 mm downstream. Fan voltage is varied to control the coflow velocity, which is measured with a hot-wire anemometer between the inlet honeycomb and screen. This anemometer was calibrated in normal gravity using a hot-wire anemometer 36 mm downstream of the duct entrance. Fuel flow rate was adjusted manually and controlled with a mass flow meter. The flames were ignited with a hot-wire ignitor. Video was recorded with an analog color video camera and downlinked in real time. Still images were recorded with a 12 bit Nikon™ D100 single-lens reflex color digital still camera (with 3008 × 2000 pixels) with a 60 mm lens.

Most of the tests were conducted by setting the coflow velocity, igniting the flame, and adjusting the fuel flow rate. Approximate smoke points were identified in the video by the flight crew and simultaneously by the ground support crew in Cleveland. Typical burn times were 60 s, and after each two such burns the MSG contents were flushed with ISS cabin air.

After the tests the video record was analyzed to identify smoke point conditions. The associated coflow and fuel flow rates were obtained from digital data recorded in the video. The smoke point lengths, L_{SP}, were obtained from the video record and are the measured distance from the burner tip to the tip of the luminous flame on the flame axis. The luminous flame tips were not sharply defined in most images, but rather involved a transition from bright yellow to black across an axial distance of 2 – 5 mm. Flame height was thus measured to the point where the centerline grayscale intensity changed most rapidly between these yellow and black regions. This helped reduce the sensitivity of measured flame height to camera exposure settings.

Smoke points were obtained for five fuels, as summarized in Table 1. Smoke points were

attempted with methane, but could not be obtained before the flames impinged on the plate at the end of the duct. Also shown in Table 1 are the ranges of various parameters for which microgravity smoke points were observed.

Table 1. Summary of smoke points observed.[a]

Parameter	C_2H_4	C_3H_8	C_3H_6	75% C_3H_6	50% C_3H_6
Z_{st}	0.0637	0.0603	0.0637	0.0768	0.1018
T_{ad}, K	2367	2264	2332	2317	2288
d, mm	0.76 – 1.6	0.76 – 1.6	0.41 – 1.6	0.76 – 1.6	0.76 – 1.6
u_a, cm/s	18 – 58	16 – 45	5.4 – 57	13 – 65	11 – 49
L_{SP}, mm	37 – 105	46 – 105	13 – 59	15 – 44	22 – 50
u_f, cm/s	69.6 – 734	48.9 – 460	11.2 – 763	19.9 – 272	49.6 – 112
$\dot{m}_f$, mg/s	1.62 – 3.89	1.80 – 3.84	0.39 – 1.73	0.64 – 1.98	1.45 – 3.29
Re	127 – 636	176 – 788	36.4 – 630	49.8 – 324	89.4 – 207
# smoke points	6	6	20	13	7

[a]All smoke point flames were attached to the burner, i.e., not lifted. For mixtures the fuel mole fraction is shown and the inert is N_2. Parameters u_a, u_f, and *Re* assume conditions of 298 K, 101 kPa and 21±1% O_2, and uniform velocity profiles. Viscosities used to calculate *Re* involve burner gas properties at 298 K and 101 kPa.

3. Results and Discussion

Some smoke points, propane and ethylene in particular, were identified by the onset of gradual (rather than sharp) dimming and reddening of the luminous flame tip in the downstream direction. Rounding of the tip also was associated with the smoke point. For conditions sufficiently above the smoke point, quenched soot could be seen leaving the flame. The brightest flames did not always display significant reddening or darkening near their tips, except well above the smoke point. This occurred for the pure propylene and propylene mixtures, and the smoke point was identified by the rapid transition to an open-tipped, soot emitting flame.

The flames generally reached steady conditions within 5 s of ignition or a change in flow rate of fuel or air. Flames of propylene and propylene mixtures were generally more luminous than those of methane, ethylene, and propane, attributed to their higher propensity to form soot. The smoke point transition was more abrupt in the propylene-based fuels than for propane or ethylene. Periodic flame motion was observed in some flames, especially those with high coflow, long length, and low fuel velocity, and is attributed to non-uniform coflow. This effect was small and did not prevent clear measurement of flame length.

Figure 1 shows images for representative flames near their smoke points. Two sequences are shown, and in both the smoke point occurs between the third and fourth flames. In the top sequence, coflow velocity decreases from left to right at constant fuel flow rate. This increases luminous flame length and reduces the smoke point leading to soot emission. The luminous length decreases in the fourth image owing to incomplete combustion. In the bottom sequence fuel flow rate is increased at constant coflow velocity, increasing flame length as expected. As seen in Fig. 1, luminous length is more sensitive to changes in fuel flow rate than to changes in coflow velocity.

In Fig. 2 the measured smoke point lengths are plotted with respect to coflow velocity. For a given choice of fuel and burner diameter, an increase in coflow velocity generally increases the smoke point length. This behavior is attributed to decreased residence times in the soot formation region when coflow velocity is increased. The variation of smoke point with coflow velocity is greater for longer flames.

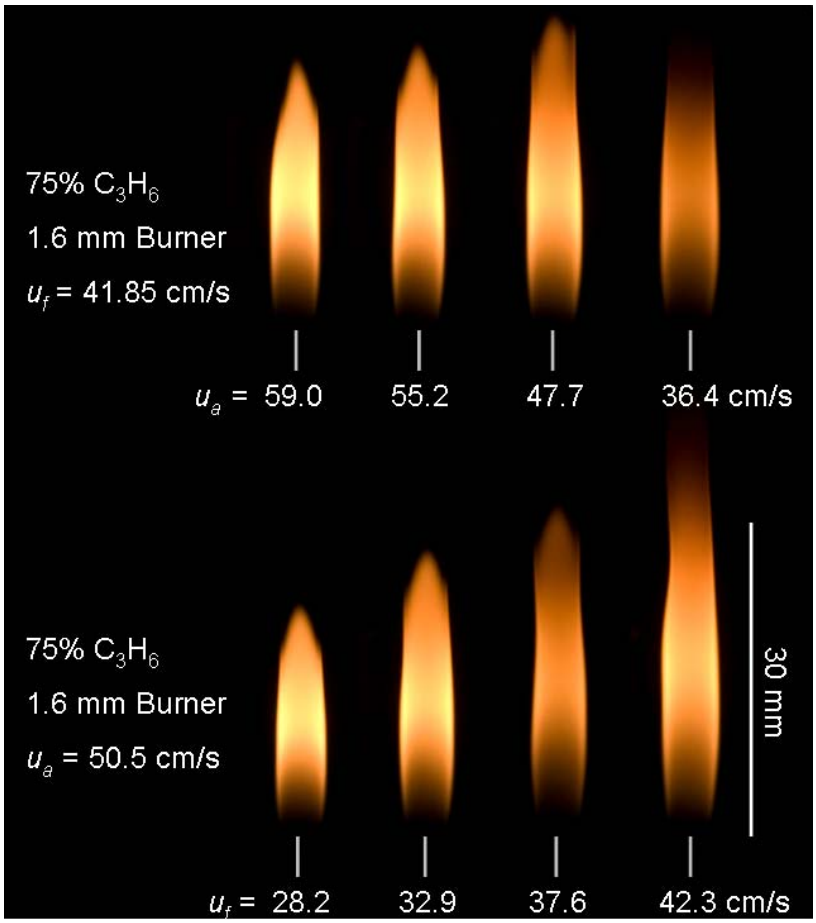


Figure 1. Color still camera images of 75% C_3H_6 flames with (top) decreasing coflow velocity at a constant fuel flow rate, and (bottom) increasing fuel flow rate at a constant coflow velocity. Images were taken at f/11 with a shutter speed of 3.1 ms (except the 4th top and 3rd bottom pictures, which involved a shutter speed of 1.3 ms).

Figure 2 reveals that smoke points of coflowing nonbuoyant flames increase with decreasing burner diameter. This differs from previous observations of microgravity flames in still air [17,18], whose smoke points are only weakly dependent on burner diameter. However, when the data of Fig. 2 are extrapolated to $u_a = 0$, a weak dependence on burner diameter results. Normal gravity smoke points also are found to have only a weak dependence on burner diameter [17,18].

The present smoke points were found to be repeatable, except for propylene with a 0.41 mm burner. Testing on different days yielded different smoke points for this combination. This is attributed to a change in the duct exit system, which became partially clogged with soot and required a higher fan voltage to obtain a given hot-wire duct air velocity reading. The lower smoke points were obtained with a relatively clogged filter and the highest smoke point was found after the filter was vacuumed clean. For all of these tests, the flow rate was set based on the hot-wire duct velocity reading, however it appears that the flow restriction in the filter changed the coflow. Other smoke points collected on these two dates did not exhibit this same discrepancy so this coflow quality effect may only affect flames of this smallest burner diameter.

The sooting propensity of the present fuels, identified from changes in smoke point length for fixed burner diameter and coflow velocity, is generally propane < ethylene < 50% propylene < 75% propylene < propylene. (This is consistent with unpublished normal gravity results in the same apparatus.) The only exception was that for the 0.76 mm burner ethylene has a longer smoke point length than propane. This could be a result of residence time effects associated with

the lower density of ethylene.

In Fig. 3 the smoke point flame lengths are plotted with respect to $\dot{m}_f / Z_{st}$. This is motivated by Eq. (3). This choice of axes yields a reasonably good single correlation of the luminous lengths of all fuels. Although this plot is for conditions at smoke points, data for flames not at their smoke points are expected to generally follow this correlation. This plot does not account for the decrease of flame length with increased coflow velocity (see Fig. 1). These results are similar to those of Lin and Faeth [25] in that the proportionality between L_{SP} and $\dot{m}_f / Z_{st}$ increases for flames with low u_a / u_f velocity ratios (see Eqs. 1 and 2). This effect on lengths is observed for propylene, propane, and ethylene flames with the smallest burner, for which $u_a / u_f < 0.08$.

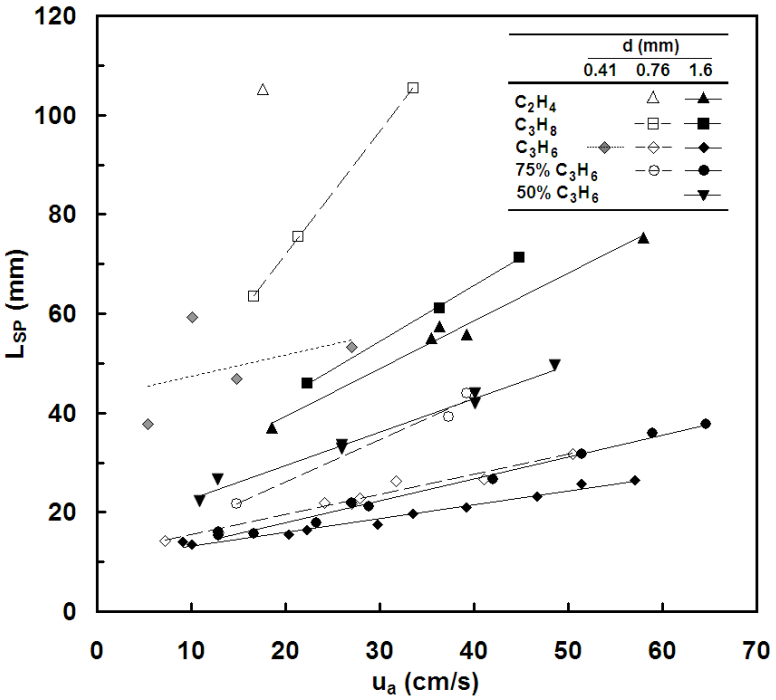


Figure 2. Smoke point luminous flame length (L_{SP}) plotted with respect to coflow velocity (u_a). The lines shown are fits for each pairing of fuel and burner diameter.

4. Conclusions

Smoke points of microgravity gas-jet diffusion flames were observed. Conditions emphasized small burners and effects of coflow velocity. These flames allow extensive control of residence time, which is important in smoke point phenomena. The major conclusions are as follows.

1. An increase in coflow velocity increases the smoke point length.
2. A decrease in burner diameter increases the smoke point length.
3. These two effects can be explained qualitatively by considering residence times; increased fuel and/or coflow velocities yield increased smoke point lengths.

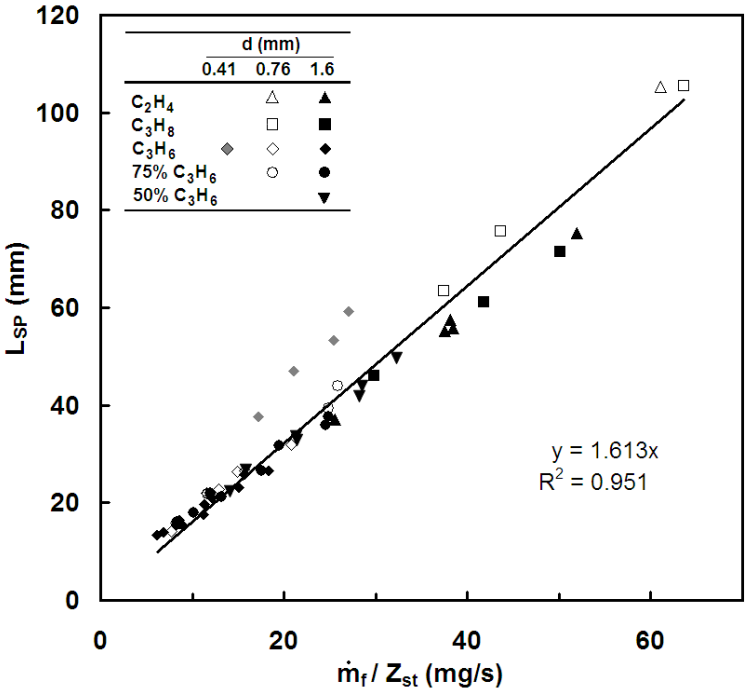


Figure 3. Smoke point luminous flame length plotted with respect to fuel mass flow rate divided by stoichiometric mixture fraction. A linear fit that intersects the origin is shown.

4. The fuel sooting propensities in these flames are propane < ethylene < 50% propylene < 75% propylene < propylene, however ethylene is less sooty than propane for the 0.76 mm burner.
5. To first order, luminous length at the smoke point is proportional to fuel mass flow rate, consistent with past theory and experiments. A second-order effect is that this length increases for low u_a / u_f velocity ratios.

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